

Grok Review — Outstanding Issues and Concerns

PascalRecon (Apple Pascal 1.3 / 128K). Branch **Grok-Review**, 2026-09-15. Omits reconstructions that already match the shipped bytes exactly. Full verification record: docs/GROK-REVIEW.md.

Accepted exception

SYSTEM.LIBRARY — code segments and interface text through IMPLEMENTATION match; the **file cannot match whole**. Text blocks came from a compiler other than the shipped one (wrong block counts, N/X trailer bytes, buffer copies). In place on disk: **16,234** differ; slot-aligned: **16,417 / 19,456**. HANDOFF §6.1: do not reopen without new evidence; a reviewer may still examine UNITPART's block-count rule if new evidence appears.

Named remainders (believed unclosable with shipped tools)

Leftover session memory or version stamps — not missing logic:

File	Bytes	Why they stay
128K.PASCAL	571	Finishing-tool slack
SETUP.CODE	3,943	Pre-version-word + stale pads
LINEFEED.CODE	467	1.1 version bits + slack
FORMATTER.CODE	394	Linker slack
LIBMAP.CODE	311	Linker slack as <i>difference</i> count; tail past SEGEND is 332 bytes (21 matching)
6502.ERRORS	287	First record-window fill
Four .MISCINFO	556	Memory SETUP never writes
BINDER / SET40COLS	16 each	SEGINFO version 2 vs 6
FORMATTER.DATA / boot	1	Uncleared assembler byte
Directory / free residue	—	Dead Filer space; APPLE2 block 187 (LINKER.INFO)

Copied, not rebuilt

SYSTEM.ASSMBLER's PASCALIO segment — nothing on the 1.3 disks rebuilds it; borrowed from Apple's file (finding 235b).

Active work: placeholder names (bytes unchanged)

- bulletTURTLEGR.TEXT — ~121 L<nnnn> labels (II.0 turtle_graphics/ is a lead)
- bulletpascalio/ and transcendental/ against those units
- bulletLeftover placeholders: Linker (G29, G91, LK2...), Librarian, LibMap (L85), Filer (~12), Editor (~24), PASCALSYSTEM (~14)
- bullet**SYSTEM.ASSMBLER** — ~204 placeholders (UCSD I.5 only; no II.0 assembler found)
- bullet**128K.APPLE** interpreter — ~800 labels; edit 128K.hints, not generated .TEXT (Brooks 1.4 is a candidate)

Probe / verification gaps

- bullet**SYSTEM.COMPIILER**: no kept-source lock in the librarian epoch (only LIBCOMP.CODE); a rename that does not move bytes would not fail that probe
- bulletFinding 290 TURTLEGR buffer fragments ({ \$endc }, path strings) are documented, not asserted by probes (LONGINTI/DECOPS is)

- bulletmkimages / SRCHD: file counts verified (35 / 73); **UTURTLE compile from SRCHD** and commented BODY13 slack still unverified under the emulator

- bulletmkimages.py / mkreport.py are outside build_all.py — delivery artifacts can drift without failing the mandatory build

- bulletFINDINGS §287 / §289 headlines still show pre-290 totals; live canon is **359,323** and library **16,417** (290d)

Environment / process risks

- bulletPort **1977** can be stolen by Windows; SYSHD left armed with SYSTEM.STARTUP looks like a dead boot

- bulletcp2 never overwrites — delete destinations first or you score a stale run

- bulletstagefile.py can refuse a moved . layout while a batch keeps compiling the **old** staged file

- bulletAppleWin must not mount evidence/ read-write (date stamps)

Disk account (context, not a failure)

Overall **359,323 / 430,080** identical; every remaining difference is named in probe_diskset.py and balanced on the last build_all run. The large in-place gap is mostly **SYSTEM.LIBRARY** plus the archived 64K pair and the remainders above.